

Che-Sang Park

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Research Interests

Robot Motion Planning with Generative Models
Physical AI, Robot Foundation Model, VLA
Reinforcement Learning, Generative Models

Education

Seoul National University <i>Ph.D. in Mechanical Engineering</i> Robotics Laboratory (https://sites.google.com/robotics.snu.ac.kr/fcp/) Advisor: Prof. Frank Chongwoo Park	Seoul, Republic of Korea <i>Sep 2020 - Present</i>
Seoul National University <i>B.S. in Mechanical Engineering</i> <i>B.S. in Computer Science and Engineering</i> Graduated <i>cum laude</i> (GPA: 3.86/4.3)	Seoul, Republic of Korea <i>Mar 2013 - Aug 2020</i> <i>Mar 2018 - Aug 2020</i>
University of Washington <i>Exchange Student</i>	Seattle, WA, USA <i>Sep 2019 - Dec 2019</i>

Publication

[P1] Spatial Attention: Adapting Execution Horizons for Diffusion Policies via Observation Sensitivity
Che-Sang Park, Junsu Ha, Jianlong Fu, Frank C. Park
arXiv, 2026

[C2] CompVLA: A Variable Compliance Vision-Language-Action Model for Contact-rich Manipulation
Jongmin Kim*, Junsu Ha*, **Che-Sang Park***, Minchang Song, Hyeokju Jeong, Himchan Hwang, Jianlong Fu, Frank C. Park
CoRL 2026

[C1] Value Gradient Sampler: Learning Invariant Value Functions for Equivariant Diffusion Sampling
Himchan Hwang, Hyeokju Jeong, Dong Kyu Shin, **Che-Sang Park**, Sehee Kweon, Sangwoong Yoon, Frank C. Park
AISTATS, 2026

[J1] Diverse Policy Learning via Random Obstacle Deployment for Zero-Shot Adaptation
Seokjin Choi*, Yonghyeon Lee*, Seungyeon Kim, **Che-Sang Park**, Himchan Hwang, Frank C. Park
IEEE Robotics and Automation Letters (RA-L), 2025

Research Experience

Microsoft Research Asia Intern (Full-time, Mentor: Jianlong Fu) Developing Robust and Efficient Diffusion Policy by applying Attention	Beijing, People’s Republic of China <i>Sep 2024 - Mar 2025</i>
SDU Elite Robot Summer School 2022 Participant Networking with other international robotics students	Odense, Denmark <i>Aug 2022</i>
Music & Audio Reseach Group Intern (Full-time) Lyric Alignment & Face Generative models using human voices	Gwanggyo, Republic of Korea <i>Jun 2019 - Aug 2019</i>
Biorobotics Laboratory Intern(Full-time) Soft robot grasping control with deep neural network	Seoul, Republic of Korea <i>Mar 2018 - Nov 2018</i>

Projects

Hierarchical Robot Foundation Model with Motion Description Language <i>Project Co-Leader</i> Co-leading of motivation definition, model structure implementation	Seoul, Republic of Korea <i>July 2025 - Present</i>
Bringing Attention into Diffusion Policy: Adaptive Execution Horizon for Efficient and Responsive Control <i>Project Leader (Advising: Jianlong Fu)</i> - Took the lead in defining the motivation, developing the core idea, and conducting experiments - Achieved a 20% performance increase over Diffusion Policy under the same computation cost	Seoul, Republic of Korea <i>Sep 2024 - Present</i>

Project Manager

- Integration of research efforts from six distinct groups into a cohesive, real-world robotic task scenario

ManiSkill Challenge (<https://sapien.ucsd.edu/challenges/maniskill2021/>)

Online

Team Leader

Jul 2021 - Jan 2022

- Lead the team to apply offline RL algorithms to 7-DOF robots to conduct complex manipulation tasks

Teaching & Advising

[M2794.00600 001] Optimization and Machine Learning	Spring 2026
Head TA	
[M3228.002300 001] Geometric Methods for High-Dimensional Data Analysis	Fall 2025
Head TA	
[M2794.009600 002] Advanced Topics in Mechanical Engineering 1 (Robot Vision)	Fall 2023
TA	
[M2794.001200 001] Dynamics	Fall 2021
TA	
[M2794.002700 001] Introduction to Robotics	Spring 2021
TA	

Skills Summary

Languages: Korean(Native), English(GRE: V 154 Q 169 W 3.0 (Mar.23.2019), TOEFL: 102/120 (Sep.29.2018))
Programming Languages: Python, ROS, C++, Java, MATLAB, Bash
Tools: Git, PyTorch, TensorFlow, Linux/Ubuntu, Vim, VSCode, Latex

Honors and Awards

BK21 Fellowship	Aug 2020 - Present
Full Semester Honor Scholarship	Aug 2014 / Mar 2015 / Aug 2015 / Aug 2018
Half Semester Honor Scholarship	Feb 2016